

Genetic Analysis Report

Report ID:	SEQ-64709DC8E41E
Generated:	2025-12-04 17:33:40
Patient Name:	Genetic Risk Analysis Report - HIGH RISK

Analysis Accuracy

93.8%

Patient Information

Analysis performed on file DNA-0EE365B99D37. Processing completed using GENO AI v2.0 Elastic Net Logistic Regression model. Model trained on gene expression data with 94.2% cross-validation accuracy. Risk probability calculated: 95.0%

Variant Information

Gene Expression Analysis Complete • Analyzed Genes: 237 • Mean Expression Level: 66.70 • Expression Variance: 91966.93 • High Expression Markers: 57 • Risk Assessment: HIGH RISK • Confidence: 95.0%

Analysis Results

{"report": {"sample_id": "SEQ-64709DC8E41E", "laboratory_user_id": "F55055", "generated_at": "2025-12-04T14:33:39.868805Z", "model_name": "geno_enet_pipeline.pkl", "model_version": "2.0", "total_genes_in_model": 237, "genes_analyzed": 237, "risk_level": "LOW", "risk_score_percent": 0.05, "risk_score_raw": 0.0004901697507330619, "risk_threshold": 75.0, "summary_title": "Genetic Risk Analysis Report - LOW RISK", "summary_text": "The genetic analysis indicates a LOW addiction risk profile based on the evaluated gene-expression markers. The model analyzed 237 genetic markers and calculated a risk probability of 0.05%. This score is below the 75% threshold, indicating lower risk.", "top_genes": [{"gene": "ENSG00000140264.21", "expression": 3092.0}, {"gene": "ENSG00000125356.7", "expression": 2347.0}, {"gene": "ENSG00000159788.20", "expression": 1376.0}, {"gene": "ENSG00000100216.6", "expression": 1327.0}, {"gene": "ENSG00000164604.14", "expression": 1115.0}, {"gene": "ENSG00000099795.7", "expression": 723.0}, {"gene": "ENSG00000110092.4", "expression": 701.0}, {"gene": "ENSG00000232388.5", "expression": 699.0}, {"gene": "ENSG00000150779.12", "expression": 699.0}, {"gene": "ENSG00000015475.19", "expression": 634.0}], "bottom_genes": [{"gene": "ENSG00000280027.1", "expression": 0.0}, {"gene": "ENSG00000131007.9", "expression": 0.0}, {"gene": "ENSG00000255233.1", "expression": 0.0}, {"gene": "ENSG00000250229.1", "expression": 0.0}, {"gene": "ENSG00000196893.3", "expression": 0.0}], "gene_stats": {"mean_expression": 66.7046, "median_expression": 0.0, "std_expression": 303.9023, "min_expression": 0.0, "max_expression": 3092.0, "genes_above_mean": 23,

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This report was generated automatically from genetic data analysis. For questions or concerns, please contact your healthcare provider.